A Quantum-State-Based Change Formalism for Radar Sensing and Potential Applications for Dual and Compact Polarimetric SAR

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***Abstract*— A quantum-state-based technique for change detection and analysis is presented. The method establishes a framework for change detection based on quantum states' entanglement concept. The approach is applicable to quantum sensing (QS) (i.e., low field intensity, photon pairs) but is contingent to feasibility of quantum implementations and measurements. The main objective in this work is to develop a new change detection technique for classical radar/synthetic aperture radar (SAR) remote sensing that is inspired by quantum illumination (QI) and exploits the underlying characteristics and mathematical analogy between quantum-states and radar scatter-field-states. The described application of QI concept to develop a quantum signal processing (QSP) algorithm for classical radar/SAR data exploitations introduces a new change detection approach for SAR multi-pol exploitations.**

***Index Terms*— Coherent change detection (CCD), compact polarimetry (CP), polarimetric change detection (PCD), quantum bit (Qubit), quantum illumination (QI), quantum sensing (QS), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) change detection and characterization for remote sensing have been extensively studied [1]–[8]. These techniques span over a wide spectrum of applications ranging from single-channel coherency to more in-depth phenomenology-based polarimetric SAR interferometry for change detection based on maximum likelihood (ML). The latter is applicable to all quad, dual, and compact-polarimetric SAR imagery [4], [9].

Recent advances in quantum technology, such as quantum sensing (QS), quantum computing (QC), and quantum signal processing (QSP) offer possibilities, new opportunities, and capabilities for remote sensing applications. For instance, sensitivity and selectivity of photons due to quantum effects (e.g., entanglement) can be utilized for advanced detection of difficult targets including those with low cross section and/or noncooperative nature. The concept of quantum illumination (QI) [10], [11] is a potent and fast-growing approach for quantum applications realizing advanced sensing for enhanced and secure detection. QI has been originally applied in the optical domain by taking advantage of source entanglement to interrogate, analyze, and detect low-reflectivity targets in presence of strong noise and bright thermal background [12]–[14]. Although entanglement is at the core of such schemes, it has been shown [11], [12], [15]–[17] that even when noise breaks the entanglement of photons traveling through the propagation channel, QI system protocols (e.g., secure communications) perform significantly better than classical coherent systems. Thus, entanglement of photons offers selectivity that can have sustained benefits even after the entanglement is broken, i.e., QI thrives in a noisy environment.

The described QI advantages can be effectively utilized in the microwave frequency domain. In fact, this spectrum is found to be ideal and more suitable (than optical) [18], [19] for target detection considering the natural presence of bright thermal noise background in the microwave frequency regime. It is, however, important to note that development of the required microwave frequency QI protocols is challenging [20]–[22]. Entangled coherent state life span or loss of idler is a limiting factor for quantum processing and sensing range. Idler is the entangled particle/photon that contains the important quantum state information (reference copy) required for quantum state coherence and phase conjugating exploitation. For longer range detection (longer return time), quality of the required quantum memory needs to be enhanced to extend idler life span and/or mitigate loss. One of the existing protocols sets the loss threshold at 3 dB [15]. Application of a standard optical-fiber quantum storage delay line translates this loss threshold to a (about) 12-km range limit. Consistent progress in quantum memory components and research, such as crystals, offers promising means to extend the operating range for remote sensing applications. Furthermore, importantly, quantum measurement (nonlinear and probabilistic in nature) and validation of state entanglement is a challenge for practical applications.

Having observed and inspired by core principles of QI and underlying interrelation and mathematical similarity with scattered field state formation, a quantum-state-based change formalism for radar/SAR change detection is proposed. The developed technique that can also be viewed in the context and conceptual framework of QSP is established in parallel and in relation to principles of quantum mechanics/QI and quantum state structure concept, e.g., measurement consistency and quantization. Hence, an equivalent quantum two-level (qubit) system model for radar/SAR scattered EM field (or state in a quantum-mechanical framework) is developed. As is known

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in the realm of QSP [23]–[30], however, the introduced formalism is not restricted by all constraints associated with quantum mechanics such as those associated with quantum measurements and entanglement evaluations. In fact, a system represented by a quantum model and analyzed/processed by quantum algorithms need not be quantum-mechanical framework [30]. In other words, associated quantum physics and QI principles are emulated in terms of QSP algorithms with some degree of freedom. Thus, (similar to QSP), the formalism here incorporates principles of quantum mechanics for advanced change detection without the need of being subjected to all restrict axioms and protocols of quantum mechanics. One should note that although the above-described change algorithm for classical radar/SAR is the prime objective here, the core formalism which is based on the concept of scattered-field state entanglement can be applied in quantum regime for quantum radar/sensing providing the quantum-mechanical axioms and constraints are satisfied, e.g., quantum-mechanical measurement, validation/verification, and evaluation of entanglement [31].

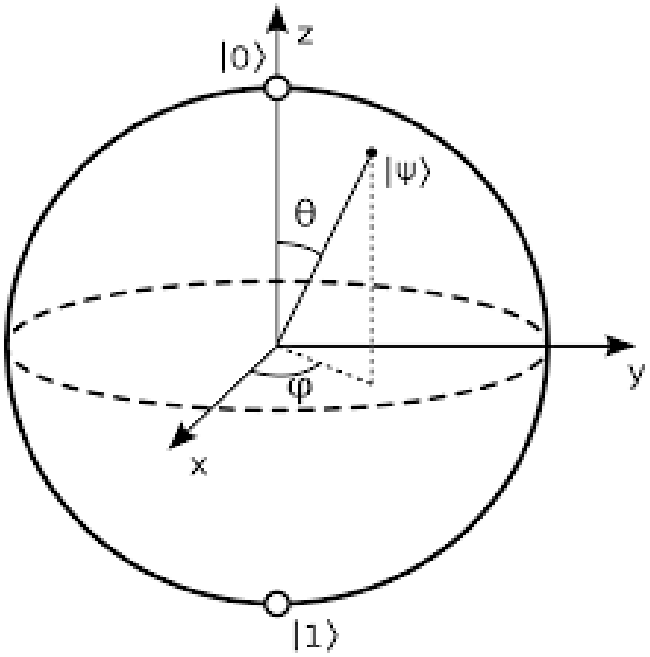
Following the above description and reasoning, a formalism for change analysis based on the "state de-entanglement" is developed and presented in the next section. It will become clear that the SAR polarimetric mode (dual and compact-pol, in particular) is a good fit for application of the developed methodology and related comparisons and/or validations. Ensuingly, change products based on the current formalism and those derived from classical techniques are explored and compared by using actual SAR dual/compact polarimetric imagery.

II. Quantum Formalism and Change Characterization

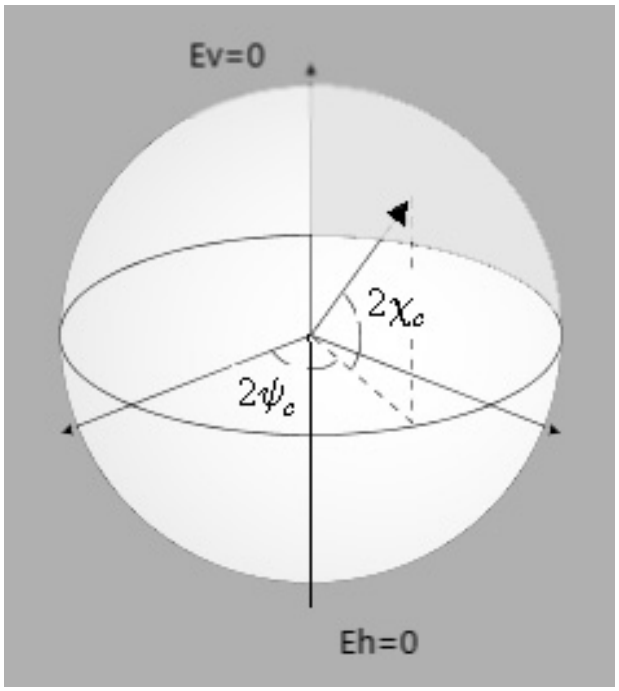
A. Quantum-State Analogy, Mapping, and Modeling for Signal Processing

One can model and represent the scattered EM field in response to a coherent illumination as a superposition state. This general field superposition can be decomposed into polarizations H and V or any other orthogonal state set. Similar to quantum state decompositions, the scattered/echoed EM signal or image by targets in radar remote sensing is, in reality, a decomposition (stochastic process) into random polarization alignments states. As is the objective of radar/SAR exploitation techniques, i.e., detection/recognition/classification, ML or probability of a response state determines the target identity.

To further explore the mathematical connection between the radar field scattering response and superposition state constructs, the analogy between the Poincare (associated with Stokes vector) and Bloch spheres (associated with binary quantum states) is depicted in Fig. 1. As is seen, there exist a homomorphism and isomorphism where the EM field $\mathbf{E}$ and superposition state $|\Psi\rangle$ are generated by scanning pair of angles χ, ψ (or more accurately $2\chi, 2\psi$) and θ, φ. The antipodal points on the spheres, i.e., top and bottom, signify the pure or determinate states $\{|0\rangle, |1\rangle\}$ and $\{|h\rangle, |v\rangle\}$, respectively.



(a)



(b)

Fig. 1. (a) Bloch Sphere representing binary quantum states. (b) Poincare Sphere representing the field Stokes Vector.

Following the above, we can represent the unified definition of the scattered field $\mathbf{E} = \begin{pmatrix} E_h \\ E_v \end{pmatrix}$, when normalized by the total field power as the superposition state in Dirac notation ($\equiv$ vector notation):

$$|\Psi_E\rangle = \alpha_h|h\rangle + \alpha_v|v\rangle \equiv \begin{pmatrix} \alpha_h \\ \alpha_v \end{pmatrix}. \tag{1}$$

This is basically state decomposition (or representation) into polarization pure or determinate states where the state

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probabilities $\{|\alpha_h|^2, |\alpha_v|^2\}$ satisfy

$$|\alpha_h|^2 + |\alpha_v|^2 = 1. \tag{2}$$

The field superposition state in (1) is similar to the Bloch sphere representation [see Fig. 1(a)] of a pure[1] qubit state given by

$$|\Psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle \equiv \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\varphi}\sin\left(\frac{\theta}{2}\right) \end{pmatrix}. \tag{3}$$

The polarization states $\{|h\rangle, |v\rangle\}$ can be generalized to any orthogonal set $\{|p1\rangle, |p2\rangle\}$ to satisfy the polarization sensing application (i.e., probe state) and objective with

$$|\alpha_{p1}|^2 + |\alpha_{p2}|^2 = 1. \tag{4}$$

The simple field presentations (1) and (2) can be used to establish a core formalism for change detection in quantum and/or classical illumination radar systems. Essentially, the above qubit state representation (one qubit/two-level for single field state, and two qubit/four-level for entangled or joint field states) prepares the quantum system model for quantum treatment of the scattered fields and required QSP for change formulation.

To draw analogy with the underlying quantum and QI principles for developing the quantum-state-based change detection algorithm, it is beneficial to first consider the QI and scattering/sensing in a quantum-mechanical framework. This quantum-mechanical framework can then be translated and transferred to a QSP framework for computations and analysis by taking advantage of the abovementioned isomorphism.

B. Quantum-Mechanical Picture

The annihilation operator $\hat{a}_r$ associated with a scattered/reflected photon (or beam mode) in response to a coherent illumination $\hat{a}_t$ operating at an RF frequency ω_R is given by

$$\hat{a}_r^{(i)} = \sqrt{\gamma_i}\hat{a}_t^{(i)} + \sqrt{1-\gamma_i}\hat{b}_N^{(i)} \tag{5}$$

where γ indicates the target reflectivity, and $\hat{b}_N$ represents overall noise and environmental modes [17]. The index (sub/super) $i = 1, 2, \ldots$ in (5) represents the sensing instant index in the change detection or analysis scenario. The return or echoed quantum signal description in (5) is similar (albeit in operator form) to those applied for classical radar/SAR scattered or return field description.

A conventional change detection and analysis scenario is constructed by application of (5) for two change interrogation instants, namely

$$\hat{a}_r^{(1)} = \sqrt{\gamma_1}\hat{a}_t^{(1)} + \sqrt{1-\gamma_1}\hat{b}_N^{(1)} \tag{6}$$

$$\hat{a}_r^{(2)} = \sqrt{\gamma_2}\hat{a}_t^{(2)} + \sqrt{1-\gamma_2}\hat{b}_N^{(2)}. \tag{7}$$

The Hamiltonian associated with the reflected/echoed photon (photon field) is given by $\hat{H} = \hbar\omega_R\hat{a}_r^\dagger\hat{a}_r$; hence, the associated statistical averaging, $\langle\hat{H}\rangle = \bar{n}_r\hbar\omega_R$, yields the average photon number $\bar{n}_r$.

Using (6) and (7), we postulate the state-based change detection as following.

1) Assuming the illumination joint state modes associated with the pair $\{\hat{a}_t^{(1)}, \hat{a}_t^{(2)}\}$ are entangled (analogous to photon and idler relation), the return states associated with the scattered or reflected fields $\{\hat{a}_r^{(1)}, \hat{a}_r^{(2)}\}$ are entangled if there is no change in target. Equally saying, if the illumination states are entangled, degree of no-change/change can be estimated by evaluating the entanglement/de-entanglement of the return states associated with $\{\hat{a}_r^{(1)}, \hat{a}_r^{(2)}\}$.

As discussed earlier and based on the extensive related literature and reported results, a QI-based sensing scheme is capable of sensitive sensing of low RCS and difficult targets in presence of bright noise. Accordingly, the above hypothesis test (in quantum-mechanical and/or QSP domain) is reasonable to evaluate complex and subtle target change. One also notes that the nature of change probed through evaluation of entanglement is more akin to physics of scattering than conventional classical evaluation of coherence.

In QS, quantum observables and information is integrated and included in the construct of the system state vector such as phase. This information is mined and extracted by applying various quantum measurement and manipulation schemes. For example, in Ramsey interferometry, the absorption or transition frequency is measured through interaction with an external field and by estimating the detuning (difference between the applied EM field and transition frequency) from the state phase. Similarly, in quantum computations, the computing target information is contained in the phase of state vector or qubit as the result of the so-called "phase kickback" (enforced by the required/designed function operator). Hence, one concludes that target reflectivity (and change) shapes the state distribution (e.g., phase) and is captured in the associated states' behavior and structure. This is true for the system states associated with the QI picture (6), (7), and radar/SAR states modeled by taking advantage of analogy and isomorphism with QI (1) and (2). Accordingly, one can state the change hypothesis as following.

1) Providing the illumination photon states is entangled [as are for conventional coherent change detections (CCDs)], the scattered states $|\Psi_{E1}\rangle$ and $|\Psi_{E2}\rangle$ are entangled if there is no target scattering change. Equally saying, the joint state associated with the scattered fields represented in Dirac form by $|\Psi_{E1,E2}\rangle$ is entangled when there is no change and separable when target reflectivity changes.

Thus, evaluation of separability and/or entanglement of the scattered joint state would result in change detection under the described paradigm. As mentioned, evaluation of entanglement for change detection in actual quantum system (see Section II-B) requires a sophisticated quantum measurement and entanglement validation/verification scheme. It is also beyond the scope of current work. Therefore, the evaluation of radar/SAR quantum-modeled states for change detection will be the focus here.

[1]Pure quantum state is a state describable by a single Dirac vector.

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C. Representation and Quantification of Entanglement for Quantum-State-Model of Radar/SAR Signals

The quantum-state-based change detection is formulized by testing the entanglement hypothesis for the quantum-modeled radar/SAR state vectors associated with the change instants. This is achieved by devising and developing a suitable QSP algorithm. We note that choice of QSP algorithm to quantify the entanglement is not unique. The proposed approach here is to evaluate and quantify the entanglement by concept of joint-state separability. Accordingly, the QSP algorithm consists of two parts: 1) construction or development of joint state associated with $|\Psi_{E1}\rangle$ and $|\Psi_{E2}\rangle$ in the form of $|\Psi_{E1,E_2}\rangle$ and 2) formulate and quantify the separability of the joint state. It is evident that each of the states $|\Psi_{E1}\rangle$ and $|\Psi_{E2}\rangle$ is equivalent to one qubit ($2^1 = 2$ level system), while the joint state $|\Psi_{E1,E_2}\rangle$ is analogs to two qubit ($2^2 = 4$ level system).

The mechanism to generate the joint state associated with radar/SAR data/image states needs some discussion and will be included in the next segment. Let us assume the joint state is created and available as a two qubit vector $|\Psi_{E1,E_1}\rangle$. To evaluate and quantify the separability of this joint state, we use Schmidt decomposition. The Schmidt decomposition that can be directly derived from standard singular value decomposition (SVD) can provide acceptable results with relative ease for applications of interest here.

It can be shown [32], [33] that for any pure quantum state $|\Psi_{A,B}\rangle$ of joint systems A and B, there exist a Schmidt decomposition to basis states/vectors

$$|\Psi_{A,B}\rangle = \sum_i \sqrt{\lambda_i}\left|u_A^{(i)}\right\rangle \otimes \left|v_B^{(i)}\right\rangle \tag{8}$$

with

$$\left\langle u_A^{(i)}\middle|u_A^{(j)}\right\rangle = \left\langle v_B^{(i)}\middle|v_B^{(j)}\right\rangle = \delta_{ij} \tag{9}$$

$$\sum_i \lambda_i = 1 \tag{10}$$

where λ_i can be viewed as probability of SVD in respective joint states. In (8) and (9), $\otimes$ and δ_{ij} represent Kronecker product and delta, respectively.

Applying the SVD method described in (8)–(10), we can quantify the entanglement [32] and [33] by defining the entropy-equivalent metric

$$H_{\text{jnt}}\big(|\Psi_{A,B}\rangle\big) = -\sum \lambda_i \log(\lambda_i). \tag{11}$$

It can be seen for separable joint states, i.e., a dominant $\lambda_i(\to 1)$, the entropy H_{jnt} is low. The entropy increases when there is entanglement. At the limit, i.e., complete entanglement, $\lambda_i = \lambda_j = \cdots$ and $H_{\text{jnt}} \to 1$.

Typical examples (using qubits) are separable state

$$|\Psi_{A,B}\rangle = |0_A\rangle|0_B\rangle \quad \text{or } U_A|0_A\rangle U_B|0_B\rangle \tag{12}$$

where U_A and U_B are any unitary transformation. For the separable state in (12), one gets $\lambda = 1$ and $H_{\text{jnt}} = 0$.

And, entangled Bell state [23]

$$|\Psi_{A,B}\rangle_{\text{Bell}} = \frac{1}{\sqrt{2}}|0_A\rangle|0_B\rangle + \frac{1}{\sqrt{2}}|1_A\rangle|1_B\rangle \tag{13}$$

where $\lambda_1 = \lambda_2 = (1/2)$ and $H_{\text{jnt}} = 1$.

The described entanglement estimator can be used for QI-modeled RF remote sensing applications. In particular, dual-polarimetric QI and sensing radars are good fit considering that the reflected EM field is already in polarization superposition state (1) and (2).

The pure or superposition states associated with the scattered RF EM fields (required for change probe) can be represented by

$$|\Psi_{Ei}\rangle = \alpha_h^{(i)}|h\rangle + \alpha_v^{(i)}|v\rangle, \quad (i = 1, 2) \tag{14}$$

where the state probabilities satisfy

$$\left(\alpha_h^{(i)}\right)^2 + \left(\alpha_v^{(i)}\right)^2 = 1, \quad (i = 1, 2). \tag{15}$$

At the receiving sensor, one can form the quantum state of joint system 1 and 2 as the pure state $|\Psi_{E1,E2}\rangle$. According to the earlier discussion (8)–(10), we have

$$|\Psi_{E1,E2}\rangle = \sqrt{\lambda_1}\left|u_{E1}^{(1)}\right\rangle \otimes \left|v_{E2}^{(1)}\right\rangle + \sqrt{\lambda_2}\left|u_{E1}^{(2)}\right\rangle \otimes \left|v_{E2}^{(2)}\right\rangle. \tag{16}$$

Decomposition factors λ_1, λ_2 are derived using the described Schmidt decomposition and SVD technique. Thus, we define the change evaluation or correlation factor

$$\text{QSchf} = -\lambda_1 \log_2(\lambda_1) - \lambda_2 \log_2(\lambda_2). \tag{17}$$

As can be seen from (17), correlation is low when the two reflected states are separable and increases when the two states become entangled. Similar to the classical CCD concept, change (as is for dark areas in CCD map) corresponds to low QSchf values.

D. Joint-State Description for Radar/SAR Applications

The formalism established through (14), (15), and (8)–(10) can estimate quantum-state-based change. First, we assume that the illuminations (transmit signals) are entangled. This is a fair assumption considering the transmit waveforms associated with CCD instants are similar or statistically correlated. Also, the scattering factor measurement implies the transmit similarity. Following this assumption, we define a joint pure state $|\Psi_{E1,E2}\rangle$ of joint system returns ($E1$, $E2$):

$$|\Psi_{E1,E2}\rangle = \langle|\Psi_{E1}\rangle|\Psi_{E2}\rangle\rangle \equiv \left\langle \begin{pmatrix} \alpha_h^{(1)} \\ \alpha_v^{(1)} \end{pmatrix} \otimes \begin{pmatrix} \alpha_h^{*(2)} \\ \alpha_v^{*(2)} \end{pmatrix} \right\rangle \tag{18}$$

where $|\Psi_{E1}\rangle$ and $|\Psi_{E2}\rangle$ are given by (14) and (15). In (18), $\langle..\rangle$ is statistical averaging that, in a quantum-mechanical context, represents averaging of state measurements, and "$*$" denotes the complex conjugate. It is highly important to note that the averaging (multilooking for SAR pixels) in (18) indicates that the joint state, in general, is not separable, i.e., cannot be expressed as Kronecker product of two states or vectors

$$|\Psi_{E1,E2}\rangle = \langle|\Psi_{E1}\rangle|\Psi_{E2}\rangle\rangle \neq \left|\Psi_1'\right\rangle\left|\Psi_2'\right\rangle$$

or

$$\left\langle \begin{pmatrix} \alpha_h^{(1)} \\ \alpha_v^{(1)} \end{pmatrix} \otimes \begin{pmatrix} \alpha_h^{*(2)} \\ \alpha_v^{*(2)} \end{pmatrix} \right\rangle \neq \begin{pmatrix} \alpha_h^{(1)} \\ \alpha_v^{(1)} \end{pmatrix} \otimes \begin{pmatrix} \alpha_h'^{*(2)} \\ \alpha_v'^{*(2)} \end{pmatrix}. \tag{19}$$

Existence of the separate states/vectors (denoted by prime superscript) in (19) indicates complete decorrelation. As was

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described in Section II-C, the adopted SVD approach is capable of decomposing a (generally) nonseparable joint state or matrix in (19), and hence, evaluating degree of entanglement/separability.

In the present quantum-modeled radar/SAR state paradigm, one can view the multilook averaging (18) and (19) of return states (containing important speckle development[2] information) as an equivalent and/or estimate of joint-state formation in QSP domain. One direct analogy to consider for this joint-state formation is the fact that multilook averaging of SAR imagery to generate covariance, in a way, emulates the equivalent SAR quantum-state entanglement (in a QSP framework). Consequently, it seems reasonable to estimate joint state by state covariance and multilooking if the quantum-state modeling of the SAR signals is valid. One notes that entanglement/separability has a deeper foundation than the cross correlation and simple projection (i.e., rank-one measurement [30]). Here, the change hypothesis is not solely examined based on cross correlation but also exploits and insists the separability concept (8)–(17).

As described, the joint state $|\Psi_{E1,E2}\rangle$ is a four-level state (4×1 vector). Using linear algebra, this vector can be uniquely mapped to a mixed-state matrix by

$$f : \vec{V}_{m.n\times 1} \rightarrow \mathbf{M}_{m\times n}. \tag{20}$$

Accordingly, the change estimate problem turns to SVD solution for the matrix

$$\mathbf{M}_{2\times 2} = \left\langle f\left(|\Psi_{E1,E2}\rangle\right)\right\rangle. \tag{21}$$

Application of (8)–(17) and $\{\lambda_1, \lambda_2, \}$, i.e., SVD solution of (21), yields the estimated change.

The quantum-state-based change detection computation and numerical implementation can be summarized in the following simple steps.

1) Form the quantum-equivalent mixed-state matrix [according to (18)–(21)]

$$\mathbf{M}_{2\times 2} = \left\langle \vec{\mathbf{e}}_1 \vec{\mathbf{e}}_2^{\dagger} \right\rangle \tag{22}$$

where $\left\{\vec{\mathbf{e}}_1 = \begin{pmatrix} \alpha_h^{(1)} \\ \alpha_v^{(1)} \end{pmatrix}, \vec{\mathbf{e}}_2 = \begin{pmatrix} \alpha_h^{(2)} \\ \alpha_v^{(2)} \end{pmatrix}\right\}$ are the normalized vectors (states) associated with the scattered fields/signals of two change states or instants 1 and 2, respectively.

2) Evaluate the separability (8)–(11) by implementing SVD of (22), as follows:

$$\mathbf{M}_{2\times 2} = \mathbf{U} \sum \mathbf{V}^* \tag{23}$$

where diagonal matrix $\sum$ contains singular values of $\mathbf{M}$ or $\sqrt{\lambda_i}$, and columns of $\mathbf{U}, \mathbf{V}$ are the left and right singular vectors of $\mathbf{M}$. Any SVD or Schmidt decomposition computational package can be used for $\sqrt{\lambda_i}$ solution of (23) and, hence, the change metric (17).

[2]Speckle is generated as the result of coherent addition of multiple returns from scene or image pixels. Although viewed as noise degrading the appearance of image, speckle contains important system, target, and image processing information.

III. Experimental and Numerical Results

The described change detection for QS needs to be explored through experiments and observations using a quantum system, i.e., QI and sensing. To serve the main objective here, i.e., quantum-based change detection for classical radar, we examine the introduced change concept by using radar/SAR coherent data.

Suitable polarimetric SAR modes for modeling the reflected EM fields (1) and (2) are conventional dual or compact polarimetric data and image products. Results from the quantum-based change formulation here can be compared with various flavors of change detection from existing techniques [1]–[9]. This serves two important purposes. First, it provides insights to potential advantages offered by the current change evaluation approach. Second (most importantly), a relatively fair agreement between results would support the discussed state assumption and/or modeling for evaluation of entanglement or change.

A. Classical Change Detection Techniques

There exists a variety of change detection methods for dual and compact polarimetry (CP). These methods can range from single channel change detection [1]–[3], [6], [7], to formalisms based on field scattering (multichannel) change or ML-based hypothesis test [4], [5], [8], [9]. All techniques, however, incorporate the scattered field covariance to formulate the change hypothesis.

One drawback of coherence (or covariance in general) as a direct measure for change evaluation is the susceptibility to error due to signal strength. Even if the statistical coherence exists, the computed coherence can be poor due to the signal imbalance. As has been investigated [6], [7], there is a need for more comprehensive treatment and exploitation of coherence/covariance for change formulation. In other words, although coherent cross correlation products contain rich information, suitable and more sophisticated application of such products is needed for change hypothesis formalism, e.g., [6], [7], ML-based algorithms. The current approach aims to satisfy the abovementioned objective by exploiting the intriguing concept of entanglement/separability for quantum-equivalent (QSP domain) correlation/covariance.

Single-channel CCD is a conventional method applied to evaluate coherent correlation/change using

$$\text{CCD} = \frac{\langle S_1^* \cdot S_2 \rangle}{\sqrt{\langle S_1^* \cdot S_1 \rangle \langle S_2^* \cdot S_2 \rangle}} = \gamma\, e^{j\phi} \tag{24}$$

where γ and ϕ are the correlation factor and interferometric phase, respectively. The scattered signals S_1, S_2 can be any of the scattering matrix components, e.g., (HH_1, HH_2), (HV_1, HV_2), (HH_1, HV_2), and (HH_2, HV_1), for horizontal (H) transmit, and so on.

Dual-Channel Change Detection treats the scattered signal as a field vector, i.e., contains polarimetric information. Such formalisms are performed by formulating the likelihoods and associated test ratios using covariance matrix. The complex Wishart [4] distribution is applied for ML change hypothesis formulation. For this scenario, the likelihood-ratio tests can

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be developed to examine the no-change H_0 (null hypothesis) versus change H_1 hypothesis. Define the multilook complex covariance matrix (2×2 for dual-pol)

$$\mathbf{X}_{ij}^{(2\times 2)} = \sum_{k=1}^{n} \vec{V}_i^k \vec{V}_j^{k\dagger} \tag{25}$$

where i, j denote the instants or pairing indices for the scattered signal/field single-look complex vectors $\vec{V}_i^k (=\mathbf{E}_i^k)$, $\vec{V}_j^k (=\mathbf{E}_j^k)$, n represents the multilook or sample number, and "†" is Hermitian conjugate. The scattered field instant samples $\mathbf{E}_{i,j}^k$ have been described earlier [see Section II-A]. In terms of scattering, the above scattered field vectors represent the associated scattering matrix components for a common transmit polarization. For instance, for an H-polarization transmit, $\vec{V} = \begin{pmatrix} V_h \\ V_v \end{pmatrix}$ elements represent HH and HV matrix elements, respectively. Similarly, for a V-polarization transmit, $\vec{V}$ elements represent VH and VV matrix elements.

The likelihood for change hypothesis or test ratio can be formulated by applying the complex Wishart distribution $W_c(2, n, \Gamma \equiv \text{Wishart Cov.})$ as follows [4], [9]:

$$Q_{ic} = 2^{4n} \left| \frac{\mathbf{X}_{ii} + \mathbf{X}_{jj}}{\mathbf{X}_{ii}^{1/2} \mathbf{X}_{jj}^{1/2}} \right|^{-2n} \tag{26}$$

where "||" stands for matrix determinant. The test ratio in (26) can be viewed as an incoherent change estimator (comparing auto-correlations) while [9]

$$Q_{cc}^{p} = \left(\frac{1}{4}\right)^{-8n} \frac{|\mathbf{X}_{ii}|^n |\mathbf{X}_{ij}|^n |\mathbf{X}_{ji}|^n |\mathbf{X}_{jj}|^n}{\left|\mathbf{X}_{ii} + e^{-i\xi}\mathbf{X}_{ij} + e^{+i\xi}\mathbf{X}_{ji} + \mathbf{X}_{jj}\right|^{4n}} \tag{27}$$

exploits the coherent change through ML test. The ambiguity phase ξ between the pair of observations needs to be estimated [9] for this analysis.

To compare with the field-state correlation, it seems beneficial to expand/generalize the CCD (single-channel projection) concept to a field vector projection. The latter resembles the hinted [Section II-D] rank-one projection. Accordingly, we introduce the complex vector CCD metric

$$\text{VecCCD} = \frac{\text{trace}\left(\mathbf{X}_{ij}\right)}{\text{trace}(\mathbf{X}_{ii})^{\frac{1}{2}} \text{trace}\left(\mathbf{X}_{jj}\right)^{\frac{1}{2}}}. \tag{28}$$

Change detection results using the above classical methods and the quantum-state-modeled technique are produced and compared in the remainder of this section.

B. Numerical and Experimental Validation: Classical and Quantum-Based Results and Comparisons

A coherent pair of SAR dual-pol (*HH-HV*) imagery collected by RARADRSAT Constellation Mission (RCM) over Pond Inlet in Nunavut, Canada, is used for simulations, validations, and comparisons. The three satellite constellation of RCM offers a four-day revisit time with a fairly fine resolution for change detection. In particular, RCM-2 (March 10, 2020) and RCM-3 (March 14, 2020) dual-pol images are applied for change analysis. The incidence angle for all images is

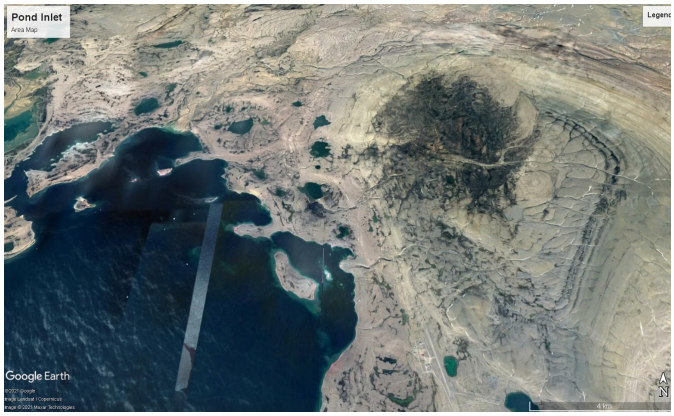


Fig. 2. Google Earth view of the area of interest AOI: Pond Inlet, Nunavut, Canada (2020). Google Earth Image: Landsat/Copernicus, Image© 2021 Maxar Technologies.

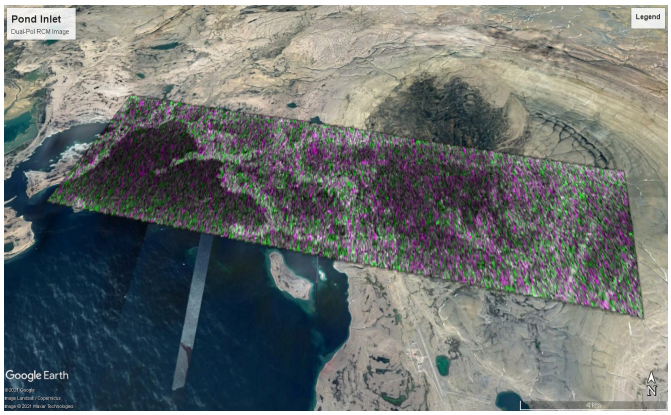


Fig. 3. RCM dual-pol image overlay of the AOI. RADARSAT Constellation Mission Imagery© Government of Canada (2020). RADARSAT is an official mark of the Canadian Space Agency. Google Earth Image: Landsat/Copernicus, Image© 2021 Maxar Technologies.

43.3 (near), 44.6 (far), with 3×1 (m^2) range and azimuth resolution. Fig. 2 presents the Google Earth view of the area of interest (AOI). The RCM dual-pol (DP) image overlay on the Google Earth map of the AOI is shown in Fig. 3. Considering the large size (fine resolution) of the RCM DP image, an image chip close to the Pond Inlet airport is used for simulations (see Figs. 4 and 5).

Figs. 6 and 7 include single channel CCD products for H and V returns (H transmit), i.e., HH_1-HH_2 and HV_1-HV_2 associated with RCM-2 (March 10, 2020) and RCM-3 (March 14, 2020) dual-pol images correlations (24) [5]–[9]. These change maps would essentially relate co-pol and cross-pol scattering variations. One can apply the derived CCD maps, i.e., *HH*-CCD and *HV*-CCD to characterize the nature of change. A more comprehensive change profile, however, requires incorporation of multi-pols (polarimetric data) to treat the scattered or echoed SAR signal as a vectorial field. The ML-based techniques [4], [5], [9] provide the desired change picture (26) and (27). In particular, the coherent dual/compact-polarimetric change detection product (CPCD) (27) [9] is derived and included in Fig. 8. A 5×5 boxcar sliding window is used for all statistical or multilook averaging in the present work.

As can be seen, different target scattering change aspects associated with the return polarization are highlighted by *HH*

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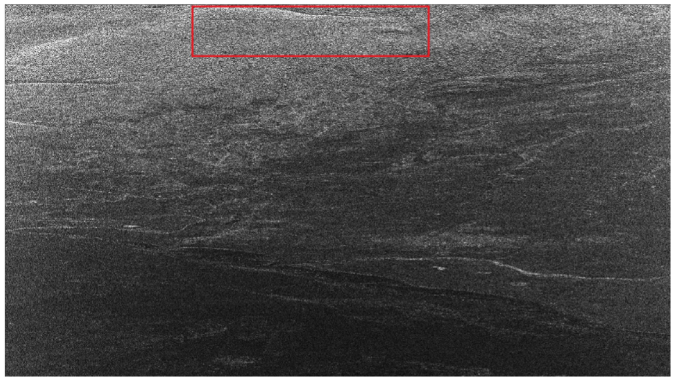


Fig. 4. Selected image area (red box) for numerical simulations and validations.

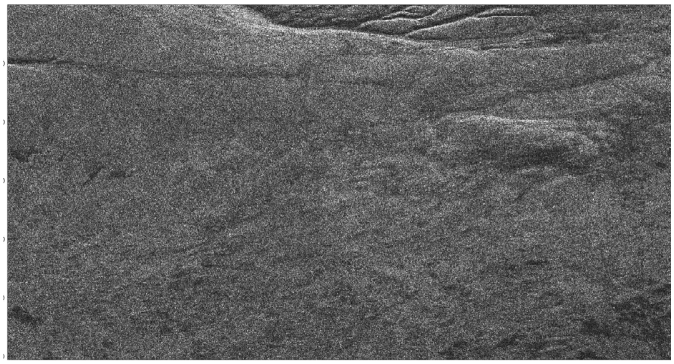


Fig. 5. Selected image chip (red box) shown in Fig. 4.



Fig. 6. Co-pol (HH_1-HH_2) CCD or correlation map associated with RCM-2 (March 10, 2020) and RCM-3 (March 14, 2020) dual-pol images of the area of interest.

and *HV* CCD profiles in Figs. 6 and 7. In particular, the co-pol return is more sensitive to capture surface scattering and associated change [8], [9], [34]. As noted though, a more comprehensive picture of change is provided by an ML-based approach characterizing the variation of EM scattering as a vector field (see Fig. 8).

Results for change detection using the introduced quantum-state-based formalism are depicted in Fig. 9. Similar box-car averaging/multilooking is applied to the joint state (18) and (22) for comparison consistency. The SVD MATLAB package is used for the required decomposition. We should note that the currently available MATLAB SVD algorithm (16), (21), and (22) introduces constraints on image chip size due to required computation time and memory. The associated change map, i.e., QSchf metric (17), indicates an overall



Fig. 7. Cross-pol (HV_1-HV_2) CCD or correlation map associated with RCM-2 (March 10, 2020) and RCM-3 (March 14, 2020) dual-pol images of the area of interest.

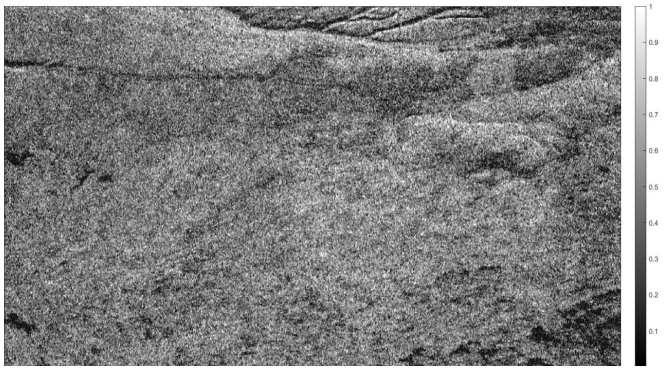


Fig. 8. Coherent CPCD (ML-based) [9] map associated with RCM-2 (March 10, 2020) and RCM-3 (March 14, 2020) dual-pol images of the area of interest.

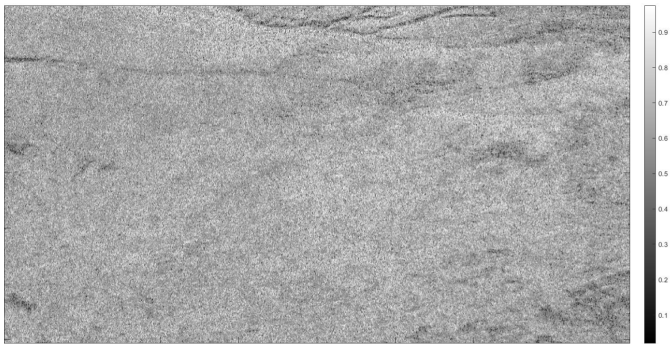


Fig. 9. Quantum-State-based change map (QSchf metric) associated with RCM-2 (March 10, 2020) and RCM-3 (March 14, 2020) dual-pol images of the area of interest.

agreement with CCD and CPCD (in particular) maps (see Figs. 6–8). Subtle differences can be analyzed by incorporating nature of the targets or scenes under investigation. Nevertheless, the results point to validity of the described quantum state assumption (mathematical) and behavior.

For a more detailed comparison, the three change maps, i.e., ML-based CPCD (ML-CCD), Vector CCD (VecCCD), and the quantum-state-based (QuantCCD) are combined in Fig. 10. The marked chips/boxes indicate regions where the lack of coherence is likely caused by low-signal levels rather than the actual scattering change. Nevertheless, the change in scattering is picked up by coherence maps. As is evident, the higher order polarimetric approaches, i.e., ML-CCD and QuantCCD

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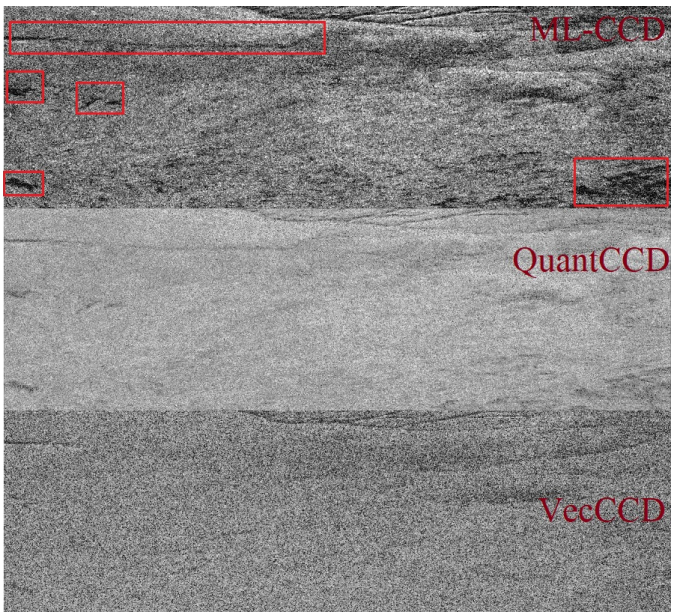


Fig. 10. Coherence maps: ML-CCD (top), QuantCCD (middle), and VecCCD (bottom) for change detection comparison. The boxed chips indicate low coherence areas that are more likely caused by low signal levels.

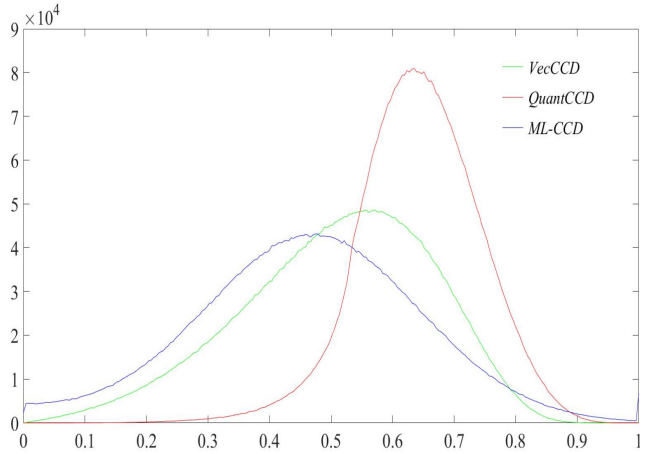


Fig. 11. Histogram envelops of ML-CCD, QuantCCD, and VecCCD correlation maps (depicted in Fig. 10).

highlight more delicate change features compared with rank-one projection VecCCD. Although experimental/numerical comparison of change algorithms is challenging, can be ambiguous (due to many factors involved), and depends on multilevel target scattering and system truthing, Fig. 10 indicates that the QuantCCD estimate is less prone to clutter decorrelation (milder decorrelation level) while maintaining the actual change behavior. To further quantify the comparison, the correlation histograms (envelop) for the three change maps are included in Fig. 11. In a change detection scenario, coherence (and its stability) of the background or clutter is important to extract and highlight change, e.g., tracks, target movement, and signature variations. Considering that change typically occurs in a limited portion of a scene or coherence map, i.e., dominated by stationary clutter, it is desirable for the overall correlation histogram to form or concentrate around higher values of correlation factor. The latter translates to a less dispersed histogram centered around a relatively high correlation value. The described behavior is manifested by the QuantCCD (see Fig. 11) in comparison with other classical methods. We note that although is not sufficient to rule and conclude the efficiency of a change detection method for all scenarios, the coherence histogram comparison can provide a reasonable quantifiable means.

IV. Summary and Conclusion

Quantum-based frameworks for remote sensing are advantageous and required considering the growing interests and advances in quantum technology such as sensing and computing. In particular, a quantum formalism is needed for low intensity/power EM field illumination where operations approach the quantum region and quantum effects at micro-level are present. For such scenarios, the classical coherent techniques (CCD) for modeling of the photon field and associated change become less effective to capture delicate variations. Although promising in view of the current growth of quantum technology, the state of the art and practical implementation of the described RF sensing construct and protocols are still challenging and at its infancy. One major challenge is the entanglement preservation for the incident/transmit quantum states. Notwithstanding the progress in extending the entanglement lifetime and quantum memory, preservation of entanglement for the quantum photons' coherence, i.e., temporal baseline, introduces a bottleneck. Albeit, the suitable or intended class of applications may not require longer temporal baselines for sensitive change probe and interrogations.

There has been considerable progress in the QSP domain which exploits underlying and mathematical similarity to employ quantum-mechanical principles for a variety of advanced signal and data processing applications. Considering the powerful and salient features of QI concept for sensitive detection, and advances in QSP, a quantum-state-based approach for radar change detection and analysis is developed and explored. This approach takes advantage of the similarity/analogy with quantum-mechanical states and entanglement principles for radar/SAR scattering states to postulate and evaluate change in a QSP framework. Although the described technique for a classical radar change detection scenario (i.e., quantum-modeling) is the main focus in this work, the core formalism which is based on scatter EM field state entanglement is applicable to QS applications (low photon density) contingent to feasibility of generating, sustaining, and evaluation/measurement of photon-state entanglement.

Although details and nuance of the change detection results are subject to interpretations and availability of more elaborate ground-truth information, the overall agreement (i.e., comparable and/or better performance) points to suitability of a quantum-state model for the return SAR signals. The introduced technique, i.e., joint-state formation and evaluation of separability, yields promising results in comparison with standard and classical SAR-based CCD change detection techniques. It is noted and discussed that evaluation of state entanglement may be approached by other techniques within a QSP framework. Hence, discussions are initiated on expansion for similar applications. Furthermore, potential considerations

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for actual QS can be made based on feasibility of photon state entanglement and evaluation.

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